Project Brief Email

From: dynamics-systems@horizon-aerospace.net
To: Controls Verification Consultant
Subject: Engineering Task Order: Drone Flight Controller Signal Filter

Dear Consultant,

Horizon Aerospace is refining the active flight control stabilization loops for an upcoming autonomous unmanned aerial vehicle (UAV). Feedback signals from the vehicle's onboard inertial measurement units (IMUs) contain high-frequency motor vibration noise, causing downstream flight control oscillations.

We need a MATLAB core script that simulates this noisy raw sensor input, builds a digital low-pass filtering algorithm without relying on toolbox extensions, analyzes signal attenuation characteristics, and updates performance logs.

Please deliver a clean script along with an engineering performance analysis report.

Regards,
Horizon Aerospace Systems Group